

PORTFOLIO - 2026

Tariq Akilah

MECHANICAL ENGINEERING

Third-year mechanical engineering student at UC Irvine with hands-on experience in vehicle and aerodynamic design, seeking engineering design roles in motorsport, automotive, and aerospace.



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BASED

San Jose, CA
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AVAILABILITY

Open for Summer 2027
US Citizen

Experiences

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Profile

Third-year mechanical engineering student at UC Irvine focused on vehicle and aerodynamic design, with hands-on fabrication experience from CAD through welding.

Currently an Aerodynamics Engineer on Anteater Electric Racing, UCI's Formula SAE EV team, developing the car's undertray in SolidWorks and validating it in STAR-CCM+. Previously, I was the avionics lead on a collegiate rocketry team developing a two-stage high-power rocket, ran a 244-case CFD analysis on a canard-controlled airframe during a research internship, and earned my National Association of Rocketry (NAR) Level 1 high-power rocketry certification.



Looking for a summer 2027 internship that challenges me to build on my experience in aerodynamics, vehicle dynamics, and vehicle structures.

Education

University of California, Irvine
B.S. Mechanical Engineering

Expected 12/2028
Irvine, CA

Mission College
A.S.-T Physics

Completed 06/2026
Santa Clara, CA

Cumulative GPA 3.88

Technical Skills

CAD

SolidWorks, Fusion 360, Onshape, AutoCAD, GD&T and Tolerance Analysis, Engineering Documentation and Drawings

SIMULATION

STAR-CCM+ (CFD), SimScale (CFD), OpenRocket, Meshing

FABRICATION

FDM 3D Printing, MIG Welding, Soldering, Mechanical Assembly, Engine Rebuild

PROGRAMMING

MATLAB, C / C++, Python

Undertray Aerodynamics

Anteater Electric Racing (Formula SAE)

Redesigned the undertray on Anteater Electric Racing's Formula SAE EV to raise downforce over 200% within the car's packaging and rules constraints.

TIMEFRAME	July 2026 to present
TEAM	Anteater Electric Racing, UCI Formula SAE EV
ROLE	Aerodynamics Engineer
TOOLS	SolidWorks, STAR-CCM+
STATUS	In progress - 160 N downforce achieved, targeting 200 N

OBJECTIVE

Hit the 200 N downforce target set by the team while working within the packaging space left available by the car's chassis, suspension, and powertrain designs. Clear all designs against the FSAE 2027 rules handbook, aerodynamics regulations under Section T.7.

ACTION

Diagnosed turbulent flow entering the inlet from the front wing's trailing edge creating a stagnant air patch under the car, then reprofiled the inlet and diffuser geometry in SolidWorks and validated the changes in STAR-CCM+.

Actively confirming manufacturability of the component with the composites team, while working with the front wing, rear wing, and sidepod teams, part of a 30+ member aerodynamics team, to produce a stable, effective full-car aero package.

RESULT

Undertray downforce raised from under 50 N to 160 N. Design work is ongoing, with a 200-250 N target set for the final iteration.

FIGURES

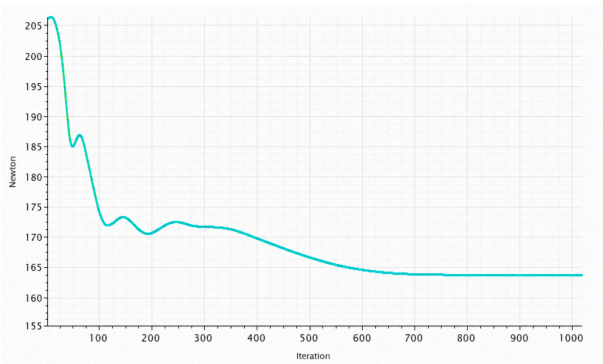


Fig. 01 - Undertray downforce convergence, STAR-CCM+

DISCLOSURE

This project is part of an active FSAE competition, so I've kept renders, CFD imagery, and design detail unpublished for now.

Canard Roll-Control CFD

Mission College Research Internship

A 244-case CFD analysis on roll torque to develop the control system for a canard-controlled rocket airframe.

TIMEFRAME	June 2026 to August 2026
TEAM	Mission College, Thrust Stack - faculty-advised research team
ROLE	Fluid Dynamics Research Intern
TOOLS	SimScale, SolidWorks
STATUS	162-point dataset delivered and adopted into the flight control logic

OBJECTIVE

Thrust Stack's airframe uses active canards to correct roll in flight, but the flight control team had no data on how much roll torque a given deflection produced at a given airspeed. Without that relationship, the controller had no way to turn a commanded correction into an actual canard angle, and the dataset had to hold up across the rocket's full flight.

ACTION

Built a SimScale sweep across the full deflection and velocity range: 244 cases, roughly 6,200 core-hours of simulation, sweeping deflection from 0.55 to 9.9 degrees and freestream velocity from 33 to 113 m/s.

The first canard geometry produced too much torque per degree of deflection for the controller to make fine corrections near small angles, so after reviewing the early sweep data, I scaled the canards down about 30% from the original to trade peak torque for finer resolution.

I also designed and manufactured a custom servo mount in SolidWorks, sized to sit flush with the airframe so the actuation hardware would not add parasitic drag.

RESULT

Converted the 162-point roll torque coefficient dataset into angular acceleration using the airframe's roll moment of inertia, then delivered it to the flight control team as the lookup table the controller reads against commanded deflection and measured rocket speed.

The conversion held up cleanly to a maximum deflection of 15 degrees. The rocket is scheduled to fly in January 2027 at the Friends of Amateur Rocketry site in the Mojave Desert, California.

Canard Actuation Demonstration: <https://www.youtube.com/shorts/M3sM7GVvhVc?feature=share>

FIGURES

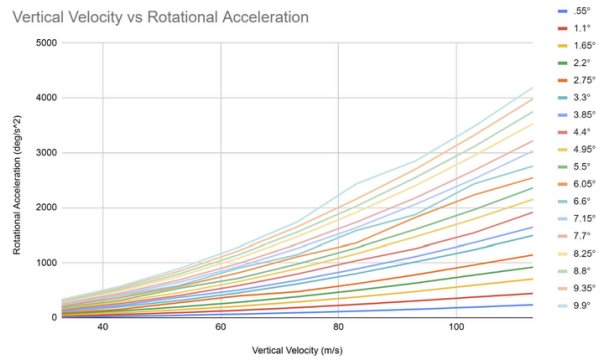


Fig. 01 - Rotational acceleration vs. velocity, by deflection

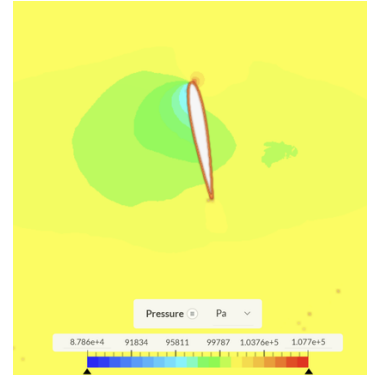


Fig. 02 - SimScale pressure field, canard section



Fig. 03 - Completed airframe

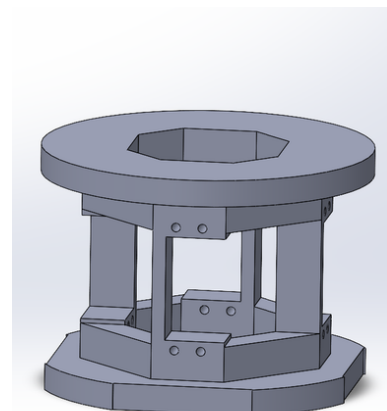


Fig. 04 - Servo Motor



Fig. 05 - Servo Fin Can

Two-Stage Rocket Avionics Bay

Mission Launch Rocketry

I led the design, integration, and testing of the avionics bay for Mission Launch Rocketry's two-stage, high-power rocket.

TIMEFRAME	July 2025 to June 2026
TEAM	Mission Launch Rocketry, five-person avionics team
ROLE	Avionics Lead
TOOLS	Onshape, SolidWorks, OpenRocket, FDM printing
STATUS	34% mass reduction, 1.12-cal stability margin, apogee predicted within 7% of measured data

OBJECTIVE

Mission Launch Rocketry set out to become the first California community college team in the Bay Area to launch a two-stage high-power rocket with dual-deploy recovery. My piece of that was the avionics bay: keep it light enough that the added mass of a second stage did not eat into our stability margin, and make the flight computers reliable enough that dual-deploy and staging worked properly.

ACTION

Designed the first iteration of the bay around fitment and structural rigidity, built to survive a hard landing without failing. After initial stress testing, I reworked the layout to shrink the bay's footprint, swapped in lighter batteries and electronics, and tuned the FDM print parameters (infill percentage, wall thickness, and infill geometry) until the mass came down without giving up structural integrity. I modeled the full airframe in OpenRocket alongside that work to confirm a lighter bay could still hold a safe stability margin once the second stage was added.

The first two launch attempts failed. Because the bay was built around dual-redundant recovery, we recovered the airframe intact both times and had flight data to diagnose the issues. The first failure came from a faulty barometric sensor reading, fixed by adding pressure relief holes to the avionics sleeve. The second was wind-induced tilt that pushed the rocket outside its staging window, so I widened the recovery system's angle tolerance to 45 degrees before the next attempt.

RESULT

The redesigned bay came in 34% lighter, worth 318 ft of simulated apogee, and the OpenRocket model held a 1.12-cal stability margin with predicted apogee within 7% of what we measured in flight. With the tolerance fix in place, the airframe is built and cleared for a final flight simulated at 5,400 ft, set to launch in September 2026.

FIGURES

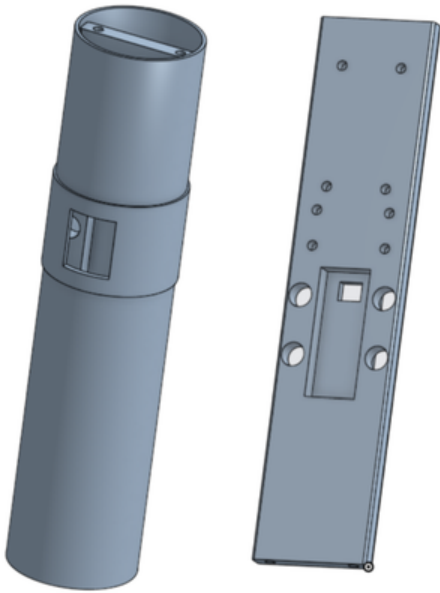
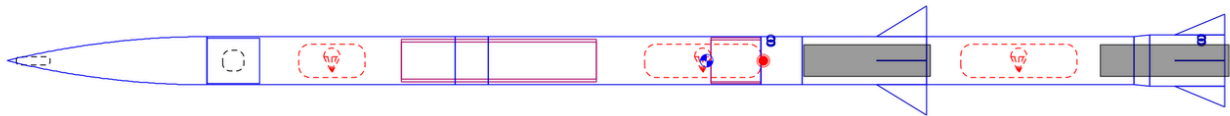


Fig. 01 - Avionics bay, Onshape model



Fig. 02 - Avionics bay, as built



Flight configuration: [None; None; H100W_DMS-14; H219T-14]
Apogee: 1650 m
Max. velocity: 201 m/s (Mach 0.596)
Max. acceleration: 16.9 G

Fig. 03 - Two-stage layout, OpenRocket



Fig. 04 - Completed airframe, for scale

High-Power Rocketry Certification

NAR Level 1 Certification

A self-designed, FDM-printed high-power rocket built, flown, and recovered to earn a NAR Level 1 certification.

TEAM	Personal build
ROLE	Designer and Fabricator
TOOLS	Fusion 360, OpenRocket, FDM (PETG-CF, Bambu Lab P1S)
STATUS	Mach 0.747 flight to 4,600 ft, recovered without critical damage - NAR Level 1 earned

OBJECTIVE

Design, manufacture, launch, and recover a rocket to earn my National Association of Rocketry (NAR) Level 1 High-Power Certification. Beyond that, I set my own performance target: keep the airframe under 600 grams without giving up the structural rigidity. The airframe had to meet all NAR high-power requirements and pass a preflight inspection from a NAR Level 3 member, and the flight itself had to end in a recovery clean enough to leave no flight-critical damage.

ACTION

Modeled the airframe in OpenRocket first, then moved into Fusion 360 to design the individual components for FDM printing. I tested a few different fin geometries and thicknesses, as well as print materials, before settling on PETG-CF, which held up under simulated flight loads without pushing the rocket over the 600 gram target.

The first launch attempt failed on recovery when the parachute detached from the airframe. I redesigned the fin can from the ground up, integrating the recovery system and motor retainer into one design, and ran the shock cord the full length of the fin can, looping it back around itself so there was no way for it to separate from the airframe short of actually cutting it.

RESULT

The redesigned recovery system held up perfectly on the second attempt. The rocket flew a stable profile to Mach 0.747 and 4,600 ft, then came down and was recovered without any flight-critical damage. That flight, combined with a pre- and post-flight inspection from a NAR Level 3 member covering structural rigidity, construction and assembly, and motor retention, earned me my NAR Level 1 High-Power Certification.

FIGURES

Mass with no motors 542 g
Mass with motors 754 g

Flight configuration: [HP-H135W-6]
Apogee: 1427 m
Max. velocity: 253 m/s (Mach 0.747)
Max. acceleration: 20.6 G

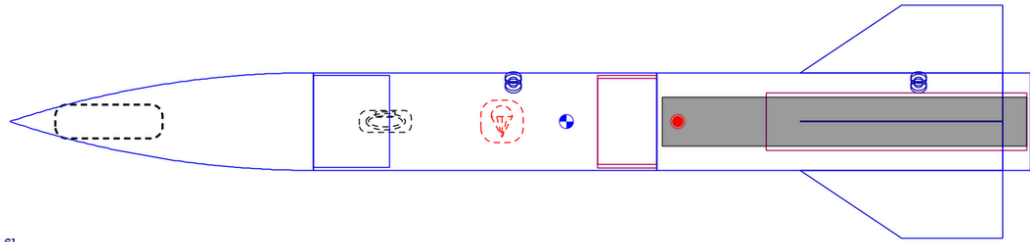


Fig. 01 - Airframe layout, OpenRocket

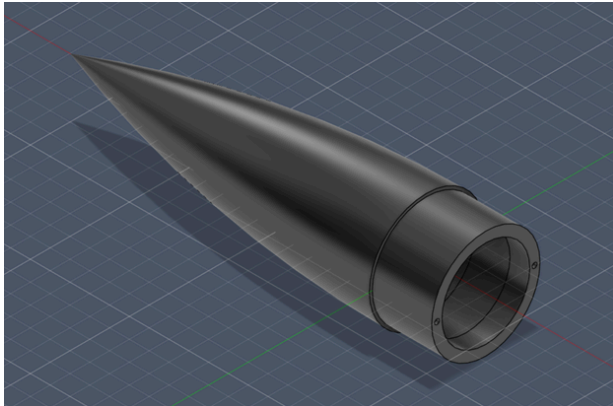


Fig. 02 - Nose cone, Fusion 360

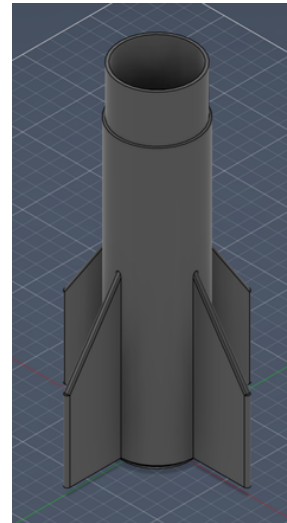


Fig. 03 - Fin can, Fusion 360

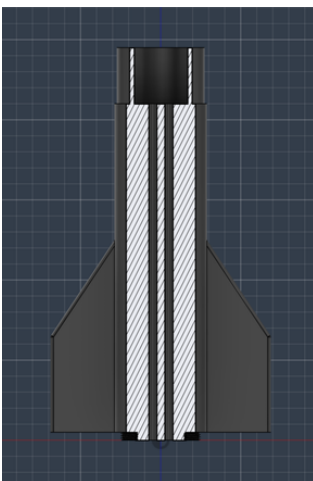


Fig. 04 - Fin can, section view



Fig. 05 - Airframe, post-flight and recovery

Motorized Trike - Full Vehicle Build

Sole Designer and Fabricator

A self-taught, ground-up motorized trike build completed within a five-week deadline and a \$500 budget.

TIMEFRAME	2024, five-week build
ROLE	Designer and Fabricator
TOOLS	Fusion 360, MIG welding
STATUS	Held a 180 lb rider with minimal to no chassis flex, 40 mph top speed

OBJECTIVE

Design and build a functioning trike within a five-week deadline, staying under a \$500 budget. I wanted to create a CAD model to reference during the build, and I also decided to teach myself MIG welding from scratch and weld the chassis myself. The chassis had to hold my body weight (170-180 lbs) without any major flex.

ACTION

Modeled the chassis in Fusion 360, then built it out of 1.75 mm wall steel square tubing, teaching myself MIG welding from scratch as I went. I had to make some design adjustments on the fly, and the welds were not the prettiest, but the chassis came together solid.

The powertrain needed more work than expected. The Predator 212 CC engine had seized, and getting it running again meant working through a clogged carburetor, inconsistent spark, and a seized flywheel. I mated the powertrain to a go-kart rear axle, and welded the chassis to a bicycle front fork for steering.

RESULT

The trike came together and ran well. It held a 180 lb rider under cruising and lateral drifting loads with minimal chassis flex. The rebuilt engine ran well and reached a top speed of 40 mph measured on Speedometer Simple, an iPhone GPS-based speedometer app. Unfortunately, I ran over on both time and budget. The wheels I had originally sourced did not fit the rear axle, so I had to order a replacement set that pushed the total roughly \$100 past the \$500 target and beyond the five-week deadline. Overall, the build met its performance objectives and was a success.

FIGURES

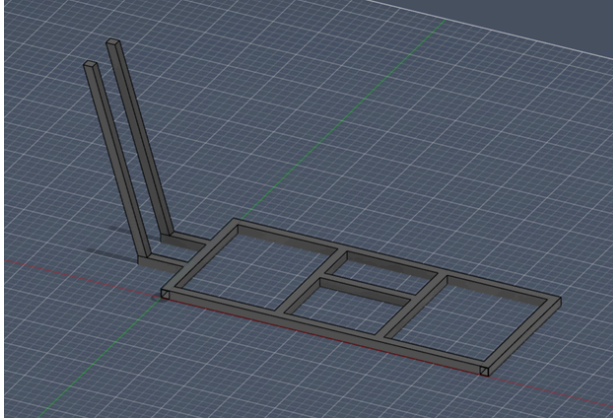


Fig. 01 - Chassis, Fusion 360 model



Fig. 02 - Chassis, rear axle fitted



Fig. 03 - Chassis, as welded



Fig. 04 - Completed trike

DAF Tank Redesign - SJSC Wastewater Facility

Team Design Project

A one-to-one replacement design for a dissolved air flotation tank at the San Jose-Santa Clara Wastewater Facility, engineered to the plant's real dimensions and delivered as a full package.

TEAM	Engineering Course Project
ROLE	Designer
TOOLS	AutoCAD, SolidWorks
STATUS	Design and package reviewed and approved by a former facility engineer

OBJECTIVE

The assignment for my engineering course was to design a one-to-one replacement for a dissolved air flotation (DAF) tank at the San Jose-Santa Clara Regional Wastewater Facility. The replacement had to drop into the existing footprint and connect to the same influent, effluent, air, and skimmer interfaces the current tank uses. I had to follow the real plant dimensions, and the deliverable had to be a complete package up to industry documentation standards, judged by a former engineer of the facility.

ACTION

I started from the facility's existing plans, pulling the tank dimensions and the locations of every plant interface so the replacement would line up without moving anything around it. From there I modeled the tank in SolidWorks and laid the civil and electrical changes into the site plans in AutoCAD.

The documentation side was as much of the project as the geometry. I helped write the addenda, contracts, and change orders to match the format a real facility project would use.

RESULT

I delivered the complete package: a SolidWorks model of the replacement tank, updated civil and electrical site plans, and the full set of contract documentation. The design and package were reviewed by a former engineer of the San Jose-Santa Clara Regional Wastewater Facility and given a full score.

FIGURES

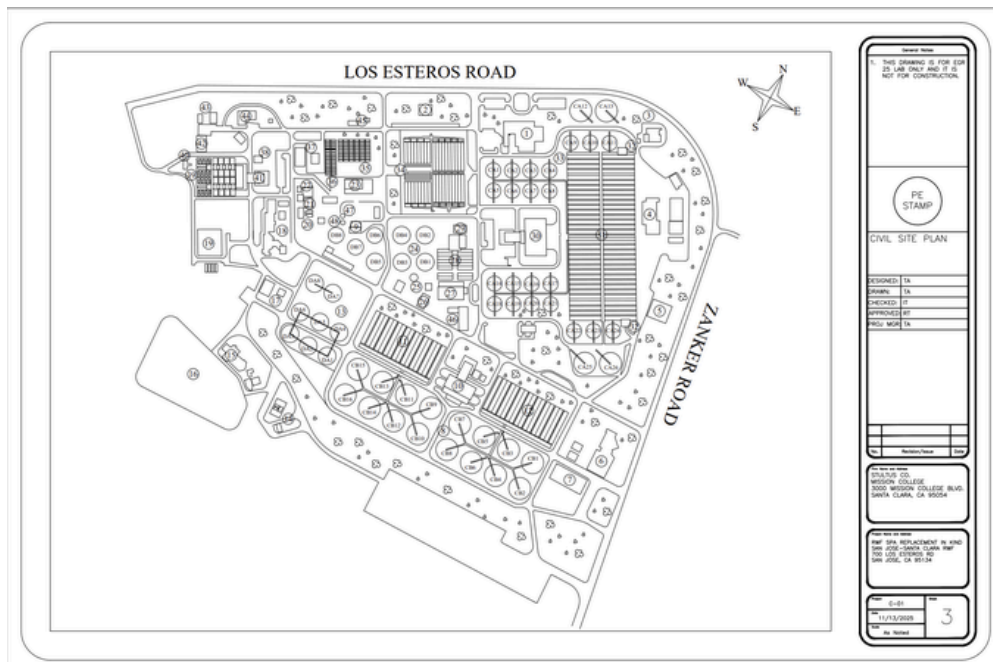


Fig. 01 - Civil site plan

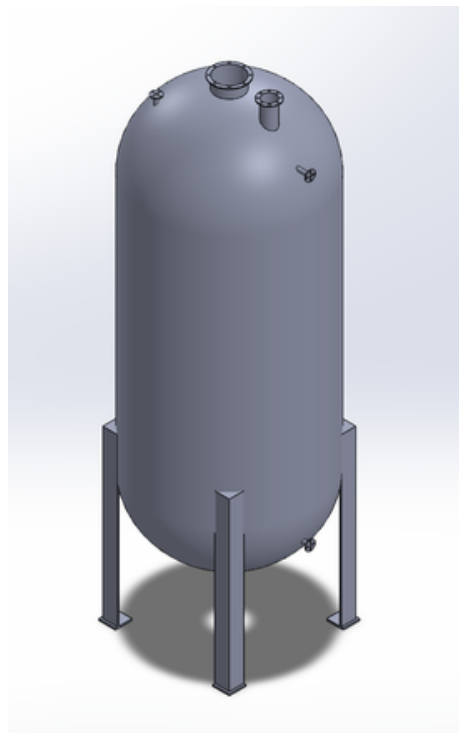


Fig. 02 - Tank model, SolidWorks

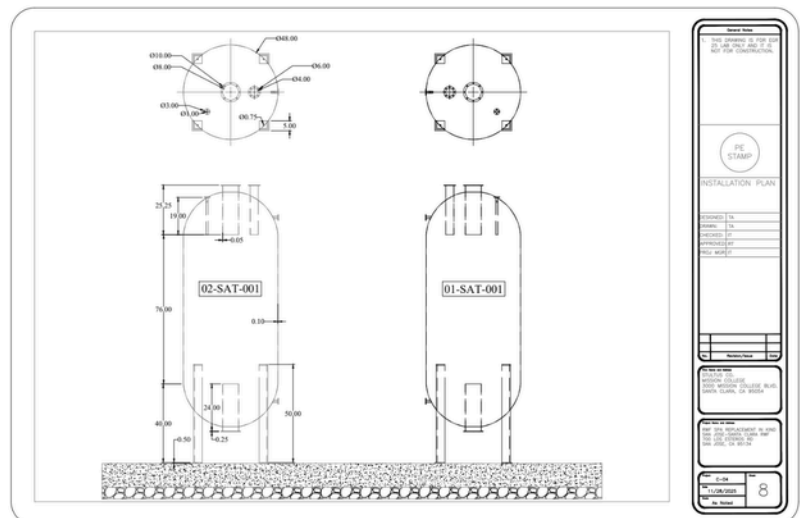


Fig. 03 - Tank drawing, AutoCAD

Coilover Suspension Assembly

Honors Engineering Project

A reverse-engineered coilover suspension assembly modeled in Fusion 360 and FDM-printed in PLA.

TEAM	Honors Engineering Project
ROLE	Designer
TOOLS	Fusion 360, PLA FDM printing
STATUS	Smooth, secure adjustment travel achieved after three revisions

OBJECTIVE

Reverse-engineer a coilover in Fusion 360 - spring, damper, shock body, adjustment ring, and mounts - then FDM-print it in PLA with tolerances tight enough to thread smoothly without seizing or rattling.

ACTION

Reverse-engineered the coilover from product photos and teardown videos, modeling it part by part in Fusion 360, then validated the range of motion with a spring-like simulation built from mates, joints, and rotational limits. My first PLA print was too tight and bound on the threads; increasing the clearance too far on the next iteration left the fit too loose, so I tuned the mating surfaces across three revisions until it moved smoothly without losing a secure fit.

RESULT

A complete Fusion 360 model and a printed PLA assembly, with an adjustment ring whose properly toleranced threads create smooth travel and a secure fit.

FIGURES

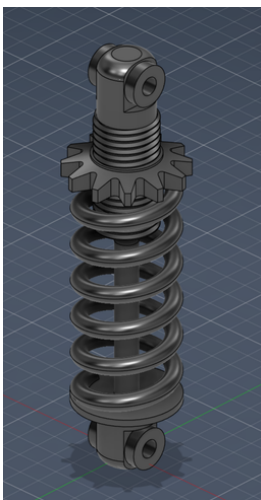


Fig. 01 - Assembly, Fusion 360



Fig. 02 - Printed components



Fig. 03 - Assembled print

CONTACT ME

Happy to talk through any of the work.

The fastest way to reach me is email. I answer everything.

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